Job Description Schema

Purpose: Standard, AI-ready structure of one job posting (Job profile) for technical and non-technical roles. The ATS matches resumes against it; the eligibility, calling and scoring engines read its per-job rules.

Files: `data/schemas/job_schema.json` (JSON Schema draft-07), `data/samples/sample_job.json`, validated with `utils/schema_validator.py`.

Design rules

- Required: job_id, company_id, title, role_category, employment_type, location, requirements (skills and minimum experience).
- Each skill has importance must_have or good_to_have so the ATS can weigh them differently.
- role_category (technical / non_technical) decides whether a machine test round applies.
- Per-job automation rules live in the job: scoring_config, interview_rounds, call_settings.
- Scoring weights should add up to 1.0 (business rule checked in Python, not in JSON Schema).
- Salary is optional because many postings do not disclose it.

Legend: Req = required (Y/-). A trailing ? on a type means the value may be null.

Top-level

Field	Type	Req	Rules
schema_version	string	Y	must be 1.0
job_id, company_id, title	string	Y	not empty
normalized_title, department, description	string?	-	
role_category	string	Y	technical, non_technical
employment_type	string	Y	full_time, part_time, internship, contract, freelance
work_mode	string?	-	onsite, remote, hybrid
location	object	Y	city, state, country (all string?)
openings	integer	-	min 1
responsibilities	array of string	-	
requirements	object	Y	see requirements
compensation, scoring_config, call_settings	objects	-	see tables below
interview_rounds	array	-	see interview_round
status	string	-	draft, open, paused, closed
posted_at	string?	-	date-time
closing_date	string?	-	date

requirements / job_skill

Field	Type	Req	Rules
requirements.skills	array of job_skill	Y	
requirements.experience.min_years	number	Y	min 0
requirements.experience.max_years	number?	-	min 0
requirements.experience.level	string	-	intern, fresher, junior, mid, senior, lead, manager
requirements.education.min_level	string?	-	secondary, higher_secondary, diploma, bachelor, master, doctorate
requirements.education.preferred_fields	array of string	-	
requirements.certifications	array of object	-	
requirements.languages	array of string	-	
job_skill.name	string	Y	not empty
job_skill.importance	string	Y	must_have, good_to_have
job_skill.normalized_name	string?	-	
job_skill.category	string	-	programming_language, framework, database, cloud_devops, tool, design, business_domain, soft_skill, other
job_skill.min_years	number?	-	min 0
job_skill.proficiency	string?	-	beginner, intermediate, advanced, expert

compensation / scoring_config

Field	Type	Req	Rules
compensation.min, max	number?	-	min 0
compensation.currency	string	-	3-letter code
compensation.period	string	-	monthly, yearly
compensation.is_disclosed	boolean	-	
scoring_config.ats_cutoff_score	number	-	0 to 100
scoring_config.weights	object	-	skills, experience, education, certifications (each 0 to 1)

interview_round / call_settings

Field	Type	Req	Rules
round_number	integer	Y	min 1
round_type	string	Y	ats, ai_screening, hr, technical, machine_test, final
cutoff_score	number?	-	0 to 100
weightage	number?	-	0 to 1
call_settings.voice_gender	string	-	male, female
call_settings.language	string	-	
call_settings.call_window_start, call_window_end	string	-	HH:MM
call_settings.max_retries	integer	-	min 0
call_settings.retry_after_minutes	integer	-	min 1